

Architecture.md Implementation

```
1 | RAG-Based Multimodal Chat Test Case Generator – Architecture
2
3 ## 1. Overview
4
5 A chat-based application that generates software test cases (Banking / Insurance domains) using a Simple/Naive RAG
6 pipeline over text documents and images. Built entirely on Langchain primitives: Model Factory, Prompt Factory, MongoDB
7 Atlas Vector Search (as a LangChain Tool), Buffer Memory, and a linear LCEL (pipe) chain.
8
9 Confirmed decisions (from requirements Q&A):
10
11 | # | Decision Area | Choice |
12 |---|---|---|
13 | 1 | Input type | Multimodal – text docs + images |
14 | 2 | Test case output format | Gherkin / Structured / JSON – user selects in UI |
15 | 3 | Persistence & export | MongoDB storage + Excel/Word export |
16 | 4 | Vision provider strategy | Try Groq vision first → fallback to configured provider |
17 | 5 | Image → RAG processing | OCR + vision description combined, then embedded |
18 | 6 | GIF handling | Extract multiple key frames for context |
19 | 7 | Vector index setup | Auto-created via startup script |
20 | 8 | Text splitter defaults | Configurable via '.env', default = medium (1000/100) |
21 | 9 | Max file size | Configurable via '.env', default = 10 MB |
22 | 10 | Chat memory | In-memory 'BufferMemory' for MVP (MongoDB-backed planned later) |
23 | 11 | Prompt templates | Hardcoded in backend config files |
24 | 12 | Domain selection | Dropdown per chat session in UI |
25 | 13 | Frontend scope | Full – Chat + Docs + Test Cases + Admin panel |
26 | 14 | Auth | None – single-user local MVP |
```

Test Case RAG Chat Documents Test Cases Admin Domain: Banking

Chat

Domain: banking New Session

Ask a question to generate a test case grounded in your uploaded banking documents.

Gherkin Attach image

Ask about banking test cases... Send

Chat

Domain: banking - session active

New Session

You

Generate the testcases for generation UPI Pin

Assistant

```gherkin  
Feature: UPI Pin Generation  
As a bank customer  
I want to be able to generate a UPI pin  
So that I can securely authenticate my transactions  
  
Scenario: Successful UPI Pin Generation  
Given I am a registered customer with a valid debit card  
When I attempt to generate a UPI pin with valid credentials  
Then I should receive a successful UPI pin generation message  
And I should be able to use the generated UPI pin for transactions  
  
Scenario: UPI Pin Generation with Invalid Credentials  
Given I am a registered customer with an invalid debit card

Gherkin

Attach image

Ask about banking test cases...

Send

## Chat

Domain: banking - session active

New Session

1. Open the net banking website.  
2. Enter valid username and an incorrect password repeatedly until the lockout threshold is met.  
3. Attempt to login again after the threshold is met.  
- \*\*Expected Result\*\*: User is locked out and cannot login. An appropriate error message is displayed.  
- \*\*Priority\*\*: Medium  
  
4. \*\*Title\*\*: Forgotten Password Recovery  
- \*\*Preconditions\*\*: User has forgotten their password and needs to recover it.  
- \*\*Steps\*\*:  
1. Open the net banking website.  
2. Click on the forgotten password link.  
3. Follow the password recovery process.  
- \*\*Expected Result\*\*: User can successfully recover their password and login.  
- \*\*Priority\*\*: High  
  
Again, please note these are hypothetical examples and not based on the provided context. For accurate and domain-realistic test cases, more specific information about the net banking login system is required.

Save as Test Case

Structured

Attach image

Ask about banking test cases...

Send

## Chat

Domain: banking - session active

New Session

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2. Enter valid username and an incorrect password repeatedly until the lockout threshold is met.  
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- \*\*Preconditions\*\*: User has forgotten their password and needs to recover it.  
- \*\*Steps\*\*:  
1. Open the net banking website.  
2. Click on the forgotten password link.  
3. Follow the password recovery process.  
- \*\*Expected Result\*\*: User can successfully recover their password and login.  
- \*\*Priority\*\*: High  
  
Again, please note these are hypothetical examples and not based on the provided context. For accurate and domain-realistic test cases, more specific information about the net banking login system is required.

Saved as test case

Structured

Attach image

Ask about banking test cases...

Send

Chat

Domain: banking - session active

New Session

```
{
 "title": "Successful Login",
 "preconditions": "User has a valid username and password, and the account is active.",
 "steps": [
 "Open the net banking website.",
 "Enter valid username and password.",
 "Click the login button."
],
 "expectedResult": "User is logged in successfully and can view their account dashboard.",
 "priority": "High"
},
{
 "title": "Incorrect Password",
 "preconditions": "User has a valid username but an incorrect password.",
 "steps": [
 "Open the net banking website.",
 "Enter valid username and an incorrect password.",
 "Click the login button."
],
 "expectedResult": "An error message is displayed indicating the password is incorrect."
}
```

JSON Attach image

Ask about banking test cases... Send

Test Cases

Domain: banking

All formats

Untitled Structured Test Case - structured - 14/07/2026, 13:09:34

Export ExcelExport Word

Untitled Structured Test Case - structured - 11/07/2026, 13:31:02

Export ExcelExport Word

ATM Cash Withdrawal - gherkin - 11/07/2026, 12:50:52

Export ExcelExport Word

ATM Cash Withdrawal - gherkin - 11/07/2026, 12:46:14

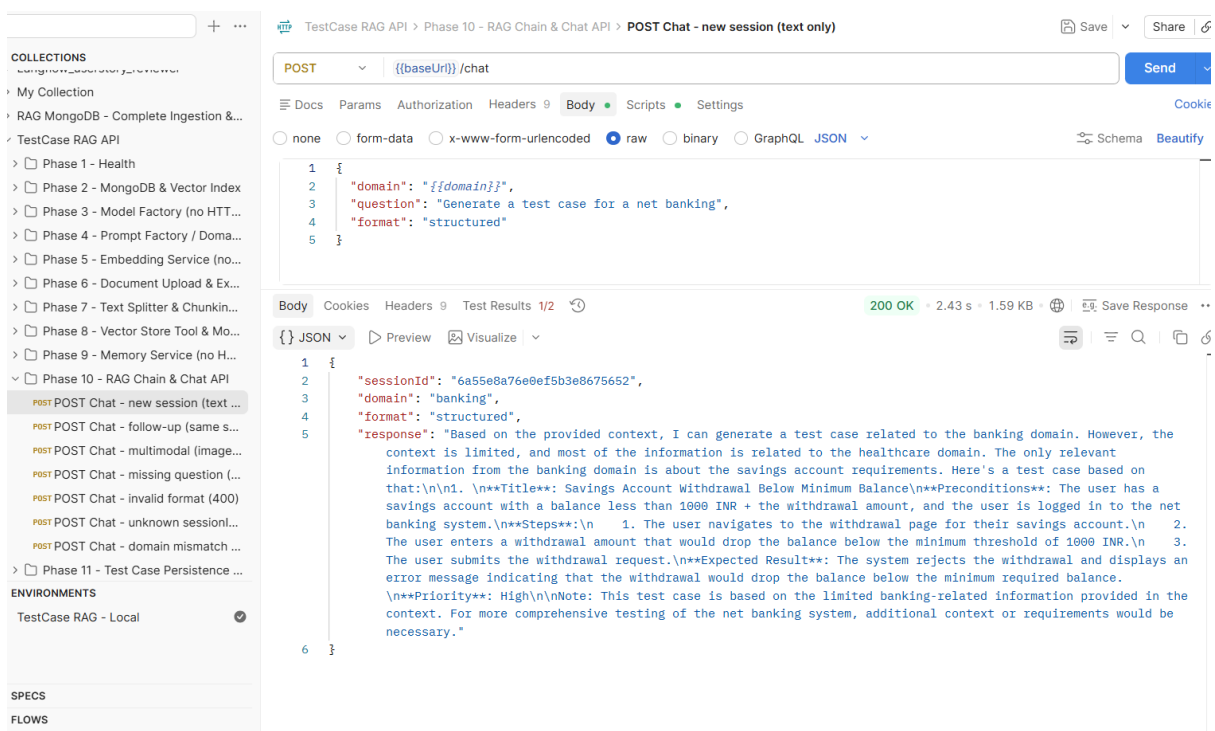
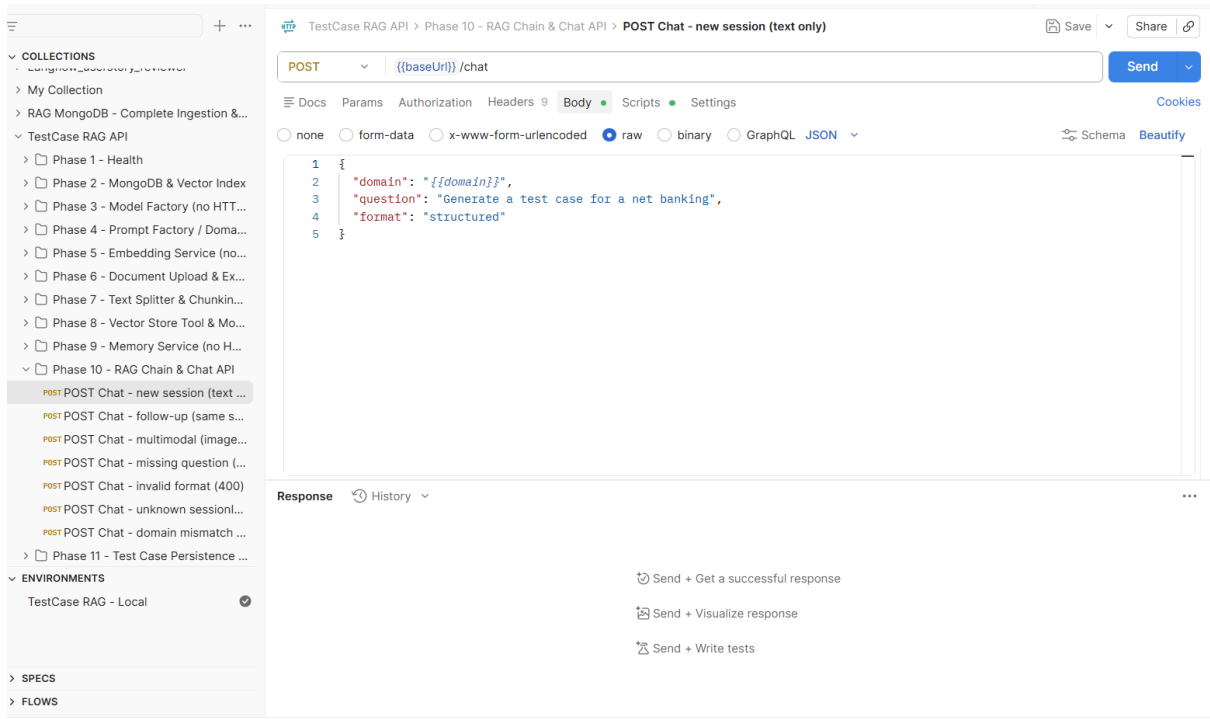
Export ExcelExport Word

Untitled Structured Test Case - structured - 11/07/2026, 12:22:37

Export ExcelExport Word

|    |                                                                          |                                                                                               |                                                                                                                                                                                               |                                                                                 |                                                                                  |   |   |   |   |   |
|----|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---|---|---|---|---|
| A1 | Untitled Structured Test Case                                            |                                                                                               |                                                                                                                                                                                               |                                                                                 |                                                                                  |   |   |   |   |   |
|    | A                                                                        | B                                                                                             | C                                                                                                                                                                                             | D                                                                               | E                                                                                | F | G | H | I | J |
| 1  | Untitled Structured Test Case                                            |                                                                                               |                                                                                                                                                                                               |                                                                                 |                                                                                  |   |   |   |   |   |
| 2  | Domain: banking   Format: structured   Created: 2026-07-14T07:39:34.243Z |                                                                                               |                                                                                                                                                                                               |                                                                                 |                                                                                  |   |   |   |   |   |
| 3  |                                                                          |                                                                                               |                                                                                                                                                                                               |                                                                                 |                                                                                  |   |   |   |   |   |
| 4  | Title                                                                    | Preconditions                                                                                 | Steps                                                                                                                                                                                         | Expected Result                                                                 | Priority                                                                         |   |   |   |   |   |
|    | Successful Login                                                         | User has a valid username and password, and the account is active.                            | 1. Open the net banking website.<br>2. Enter valid username and password.<br>3. Click the login button.                                                                                       | User is logged in successfully and can view their account dashboard.            | High                                                                             |   |   |   |   |   |
| 5  |                                                                          |                                                                                               |                                                                                                                                                                                               |                                                                                 |                                                                                  |   |   |   |   |   |
|    | Incorrect Password                                                       | User has a valid username but an incorrect password.                                          | 1. Open the net banking website.<br>2. Enter valid username and an incorrect password.<br>3. Click the login button.                                                                          | An error message is displayed indicating the password is incorrect.             | Medium                                                                           |   |   |   |   |   |
| 6  |                                                                          |                                                                                               |                                                                                                                                                                                               |                                                                                 |                                                                                  |   |   |   |   |   |
|    | Locked Out Due to Multiple Incorrect Attempts                            | The system has a policy to lock out users after a certain number of incorrect login attempts. | 1. Open the net banking website.<br>2. Enter valid username and an incorrect password repeatedly until the lockout threshold is met.<br>3. Attempt to login again after the threshold is met. | User is locked out and cannot login. An appropriate error message is displayed. | Medium                                                                           |   |   |   |   |   |
| 7  |                                                                          |                                                                                               |                                                                                                                                                                                               |                                                                                 |                                                                                  |   |   |   |   |   |
|    | Forgotten Password Recovery                                              | User has forgotten their password and needs to recover it.                                    | 1. Open the net banking website.<br>2. Click on the forgotten password link.<br>3. Follow the password recovery process.                                                                      | User can successfully recover their password and login.                         | High                                                                             |   |   |   |   |   |
|    |                                                                          |                                                                                               |                                                                                                                                                                                               |                                                                                 | Again, please note these are hypothetical examples and not based on the provided |   |   |   |   |   |

# Postman



## Unit test scripts

Explorer

- langchain\_architecture
  - backend
    - src
      - scripts
      - services
        - embeddingService.ts
        - exportService.ts
        - extractionService.test.ts
        - extractionService.ts
        - memoryService.ts
        - testCaseParserService.test.ts
        - testCaseParserService.ts
        - textSplitterService.ts
        - vectorStoreService.ts
      - tools
      - utils
      - app.ts
      - server.ts
      - test-fixtures
      - uploads
    - .env
    - .env.example
    - .gitignore
    - eng.traineddata
    - package-lock.json
    - package.json
    - tsconfig.json
    - vitest.config.ts
  - e2e
  - frontend
  - Outline
  - Timeline

Implement Architecture p... Architecture\_1.md TS vitest.config.ts

backend > TS vitest.config.ts > ...

```
1 import { defineConfig } from 'vitest/config';
2
3 export default defineConfig({
4 test: {
5 environment: 'node',
6 include: ['src/**/*.test.ts'],
7 },
8 });
9
```

Problems Output Debug Console Terminal Ports

PS C:\Users\Monisha\Documents\AI Supporting notes\Phase 2\week6\day2\langchain\_archi

VITE v8.1.4 ready in 253 ms

→ Local: <http://localhost:5173/>

→ Network: use --host to expose

→ press h + enter to show help